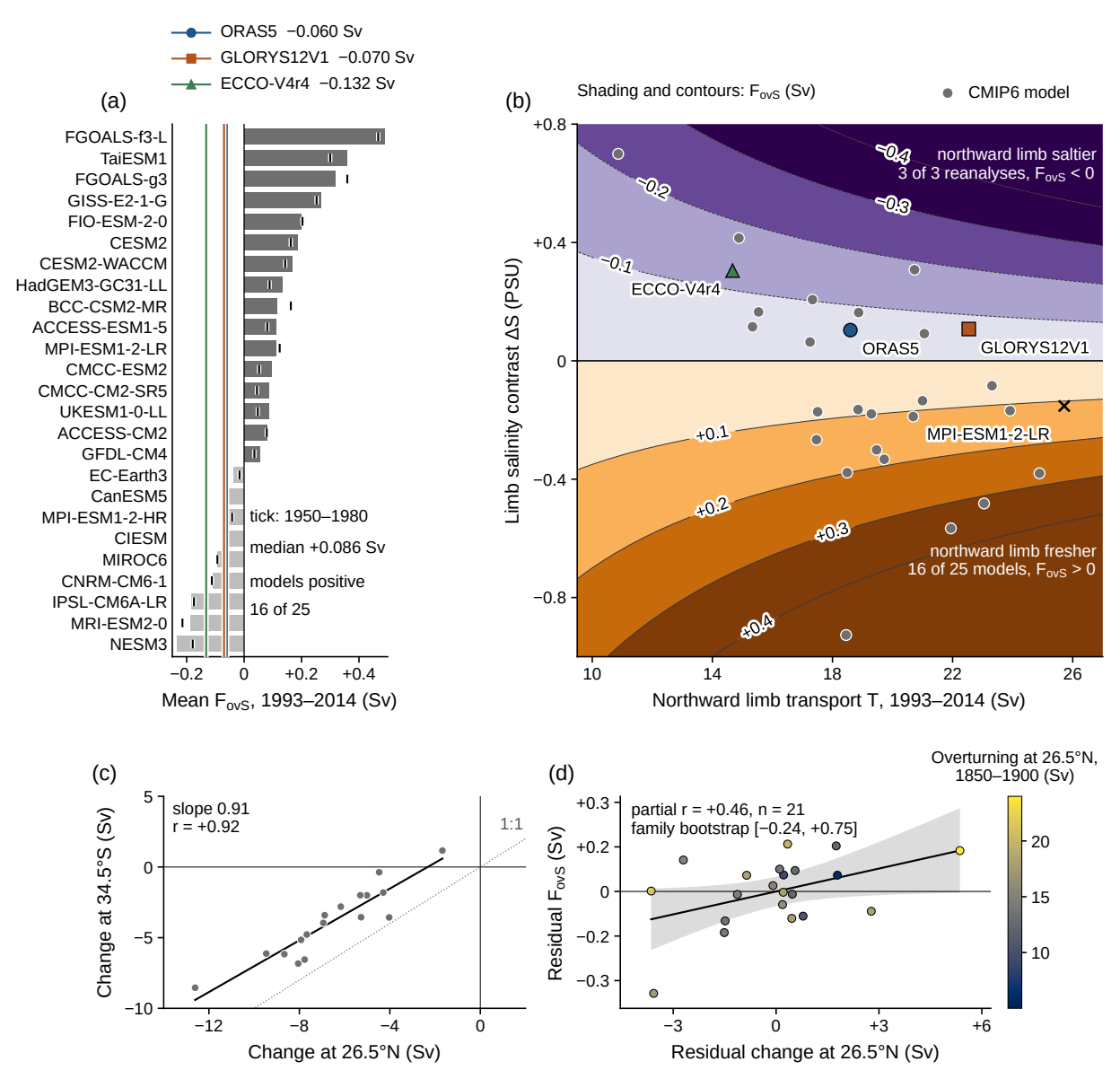

ORAS5 −0.060 Sv
GLORYS12V1 −0.070 Sv
ECCO-V4r4 −0.132 Sv
(a)
FGOALS-f3-L
TaiESM1
FGOALS-g3
GISS-E2-1-G
FIO-ESM-2-0
CESM2
CESM2-WACCM
HadGEM3-GC31-LL
BCC-CSM2-MR
ACCESS-ESM1-5
MPI-ESM1-2-LR
CMCC-ESM2
CMCC-CM2-SR5
UKESM1-0-LL
ACCESS-CM2
GFDL-CM4
EC-Earth3
CanESM5
MPI-ESM1-2-HR
CIESM
MIROC6
CNRM-CM6-1
IPSL-CM6A-LR
MRI-ESM2-0
NESM3
tick: 1950–1980
median +0.086 Sv
models positive
16 of 25
−0.2
0
+0.2
+0.4
Mean F$_{ovS}$, 1993–2014 (Sv)
(b)
Shading and contours: F$_{ovS}$ (Sv)
CMIP6 model
northward limb saltier
3 of 3 reanalyses, F$_{ovS}$ < 0
−0.4
−0.2
−0.3
−0.1
ECCO-V4r4
ORAS5
GLORYS12V1
MPI-ESM1-2-LR
+0.1
+0.2
+0.3
+0.4
northward limb fresher
16 of 25 models, F$_{ovS}$ > 0
+0.8
+0.4
0
−0.4
−0.8
Limb salinity contrast ΔS (PSU)
10
14
18
22
26
Northward limb transport T, 1993–2014 (Sv)
(c)
slope 0.91
r = +0.92
1:1
5
0
−5
−10
Change at 34.5°S (Sv)
−12
−8
−4
0
Change at 26.5°N (Sv)
(d)
Overturning at 26.5°N,
1850–1900 (Sv)
partial r = +0.46, n = 21
family bootstrap [−0.24, +0.75]
+0.3
+0.2
0
−0.2
−0.3
Residual F$_{ovS}$ (Sv)
20
15
10
−3
0
+3
+6
Residual change at 26.5°N (Sv)